

Holy Steps of Subnetting

192.168.10.0/24

31 H 15 H 6 H

11111111.11111111.11111111.11111000

STEP 1 *Determine the number of needed subnets.*

STEP 2 *Write out your host bits: 00000000*

STEP 3 *Determine the number of bits you must borrow to get the = of needed subnets.*

$2^n = \# \text{ of usable subnets}$

$n = \# \text{ of borrow bits}$

STEP 4 *Turn on (set to 1's) the number of bits you must borrow from step 3, starting with the first host bit right after the existing network bits, and find the decimal value of the octet(s).*

STEP 5 *Figure the new subnet mask by adding the decimal value from step 4 to the original (given) subnet mask (it is the classful default only when you start from a /8, /16, or /24).*

STEP 6 *Find the value of the lowest-order bit turned on in the new mask (the last bit you borrowed). This is your host subnet variable; count by it in the octet where that bit sits.*

STEP 7 *Using the variable from step 6, find the subnet range.*

Note: $2^h - 2 = \text{number of hosts per subnet}$

NETWORK	HOST RANGE	BROADCAST	HOSTS
192.168.10.0	10.1 - 10.62	10.63 /26	31 H
192.168.10.64	10.65 - 10.94	10.95 /27	15 H
192.168.10.96	10.97 - 10.102	10.103 /29	6 H
192.168.10.104			

H = number of host bits